

Student: Ioanna Pepa
Matriculation number: 01/1591952
Course: Grammar Development
Module: 4th
Lecturer: Mark-Matthias Zymla
Tutor: Kascha Kruschwitz
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A Lexical-Functional Grammar for Modern Greek

Pro-drop, Clitic Objects, and Complement Clauses

Grammar Development — Project Report

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1. Introduction and Choice of Phenomenon

This grammar implements some phenomena of Modern Greek syntax in the LFG/XLE framework. The core phenomenon is subject pro-drop which is followed by the syntactic structures that often interact with it, which are object clitic pronouns and finite complement clauses introduced by *ότι* ([`oti], 'that').

The pro-drop phenomenon (null-subject) refers to the fact that in Modern Greek a finite verb does not require an overtly expressed subject pronoun. The verb's own agreement morphology is rich enough to identify the subject on its own since it carries the person and number information, together with aspect and tense. Hence, the verb *αγαπάω* ([aɣa`pao]) unambiguously means 'I love', with no need for using the equivalent personal pronoun *εγώ* ([e`ɣo], 'I'). Choosing pro-drop as the main phenomenon brought on the surface the need to implement the other two phenomena as well because it never occurs in isolation. All the non-intransitive verbs, either they are transitive, ditransitive or optionally-transitive, when they appear in an environment where their other arguments have already been mentioned before, they can replace the full NPs of their objects with pronouns, which in Modern Greek precede the verb rather than come after it and this is the reason they are called object clitic pronouns (see examples with proper glossing in Appendix). The other phenomenon that needed to be handled was the use of complement clauses since they are very common and it would be interesting to see how the two previous phenomena work in embedded clauses that either have the same or a different subject (Philippaki-Warbuton, 1987).

2. Clause Structure: S vs Simp and OTICOMP

A grammar that only allowed overt NP subjects wouldn't represent the majority of natural Greek sentences. On the other hand, a grammar that always required a silent subject would reject the equally common cases where the subject is realized as a full NP for emphasis or disambiguation (e.g. distinguishing two third-person referents). Both possibilities are grammatical, so the grammar needed to generate both. This is the reason I decided to create two parallel top-level clause rules, one for the overt subject and one for the null-subject, which will be chosen by the ROOT rule.

```

ROOT --> { S
          | Simp }.

S -->   ({ADV*: !$(^ ADJUNCT)
        | PP*: !$(^ ADJUNCT) }) "Free word order – they can appear at the beginning of a clause"
        (NP: (^ TOPIC) = !
        (^ COMP* OBJ) = !
        (! CASE) = c acc)
        NP: (^ SUBJ) = !
        (! CASE) = c nom; "The subjects always has to be in Nominative case"
        VP: ^ = !
        (^ SUBJ PERS) = (^ VPERS)
        (^ SUBJ NUM) = (^ VNUM). "Subject-Verb agreement in person & number"

Simp --> ({ADV*: !$(^ ADJUNCT)
          | PP*: !$(^ ADJUNCT) })
          (NP: (^ TOPIC) = !
          (^ COMP* OBJ) = !
          (! CASE) = c acc)
          e: (^ SUBJ PRED) = 'pro'
             (^ SUBJ PERS) = (^ VPERS) "Pro-drop: null subject, licensed by verb agreement (person & number)"
             (^ SUBJ NUM) = (^ VNUM);
          VP: ^ = !.

OTICOMP --> OTI: @MOOD-IND; "The COMP is inserted with the use of the word 'oti'. It is either an S or a Simp"
            { Simp: ^ = !
            | S: ^ = ! }.

```

Image.1: Root, S, Simp and OTICOMP.

As we can easily see in Image.1, S requires an overt, nominative-case subject NP. Simp on the other hand introduces an empty subject (the epsilon category *e*) whose predicate is the abstract value ‘pro’, with its person and number features copied directly from the verb’s own agreement features (VPERS/VNUM), which are set by the ‘AGR’ template on each verb entry. This means that the null subject is not a guess, but its features are read straight off the verb that governs it.

Both S and Simp begin with the same optional adjunct, since both adverbs and PPs can appear in the beginning of a clause due to free word order. They are followed by an optional topicalized NP slot which is used for handling object fronting and at the end the parameters for subject-verb agreement. Complement clauses are inserted with the complementizer *óti* ([`oti], ‘that’), hence the name of the rule ‘OTICOMP’, which is always in indicative mood and can have either an S structure with an overt subject or be pro-drop shaped as a Simp.

3. Clitic Pronouns and the VP Argument Structure

Modern Greek object pronouns are clitic weak forms that attach directly before the verb. Apart from gender and number, they also carry case and they are marked with it according to whichever argument they realize, genitive for an indirect/recipient object (OBJ2), accusative for a direct object (OBJ). There is not a restriction about just one clitic before the verb. The VP rule places two independently optional object clitic slots in that fixed order immediately before the verb. This means that they can appear together but the genitive always has to come first as seen in Image.2.

```
VP --> (PRON: (^ OBJ2) = !
        (! PRON-TYPE) =c clitic
        (! CASE) =c gen)
        (PRON: (^ OBJ) = !
        (! PRON-TYPE) =c clitic
        (! CASE) =c acc)
        V: ^ = !;
        ({ NP: (^ OBJ) = ! (! CASE) =c acc;
          PP: (^ OBL-T0) = ! (! PTYPE) =c nosem
        | PP: (^ OBL-T0) = ! (! PTYPE) =c nosem;
          NP: (^ OBJ) = ! (! CASE) =c acc
        | NP: (^ OBJ) = ! (! CASE) =c acc
        | PP: (^ OBL-T0) = ! (! PTYPE) =c nosem } )
        (NP: (^ OBJ2) = ! (! CASE) =c gen)
        ({ ADV: !$(^ADJUNCT)
        | PP: !$(^ADJUNCT) (! PTYPE) =c sem } )*)
        (OTICOMP: (^ COMP) = !).
```

Image.2: VP

Some clitics are the same in both genitive and accusative, like *σας* ([sas], ‘you’-PL) and *τους* ([tus], ‘them’-MASC&NEUT). This could be very complex if each verb did not have its own subcategorization frame to decide which reading of its clitic will survive. As we can see in Image.3, I have created eight different templates according to the argument possibilities of every verb that I encountered during the development of the grammar. This way each clitic reads the potential arguments that a verb could take and according to the value of the rest of the predicates, it chooses and satisfies whichever slot matches its lexical case.

In the occasion where the verb is intransitive (INTRANS), the only argument that it will take is a SUBJ, which in Modern Greek will be dropped. If the verb is transitive (TRANS), it will take two arguments, the SUBJ and an OBJ, which is always in accusative case. The same pattern is also followed for optionally transitive verbs (OPT-TRANS). If the verb is ditransitive (DITRANS), it has two options. Either one OBJ in accusative and an OBL-TO or an OBJ in accusative and an OBJ2 in genitive. Some verbs though could also get an OBJ2 in genitive followed by a COMP (GEN-COMP) when very few can take an OBJ in accusative and a COMP (OBJ-COMP). Lastly, there is also the possibility where some of them can take just a COMP (COMP-ONLY) while others could take an OBL-TO (GOAL).

```
"Verb arguments - with examples in english for better understanding"
INTRANS(P) = (^ PRED) = 'P<(^ SUBJ)>'. "e.g. I run"
TRANS(P) = (^ PRED) = 'P<(^ SUBJ)(^ OBJ)>'. "e.g. I love chocolate"
OPT-TRANS(P) = { @(INTRANS P) | @(TRANS P) }. "e.g. I eat / I eat food"
DITRANS(P) = { (^ PRED)='P<(^ SUBJ)(^ OBJ)(^ OBL-TO)>' "e.g. I gave flowers to Mary"
               | (^ PRED)='P<(^ SUBJ)(^ OBJ)(^ OBJ2)>' }. "e.g. I gave Mary flowers"
GEN-COMP(P) = (^ PRED) = 'P<(^ SUBJ)(^ OBJ2)(^ COMP)>'. "e.g. I told you that I love you"
OBJ-COMP(P) = (^ PRED) = 'P<(^ SUBJ)(^ OBJ)(^ COMP)>'. "e.g. I informed him that dinner is ready"
COMP-ONLY(P) = (^ PRED) = 'P<(^ SUBJ)(^ COMP)>'. "e.g. I said that I love chocolate"
GOAL(P) = (^ PRED) = 'P<(^ SUBJ)(^ OBL-TO)>'. "e.g. I go to the cinema"
```

Image.3: Verb Arguments with examples in English.

4. Difficulties and Resolutions

During the development of the greek grammar I faced a few issues that worried me and took some thinking in order to overcome them. First and foremost, one of the biggest challenges was the use of the greek alphabet with XLE. I initially wanted to use my morphological analyzer that I developed for the course of Finite State Morphology, which works perfectly fine with XFST. After trying a lot of things and even creating a tokenizer by following the instructions in the book *Finite State Morphology* by Kenneth R. Beesley and Lauri Karttunen, I realised that nothing worked and I decided to continue by transcribing every greek word.

Something that caused a bit of a trouble as well is that some clitics might phonetically sound exactly the same but be different in their written form. One specific example are the clitics $\tau\eta\varsigma$ ([tis], ‘her’-GEN) and $\tau\iota\varsigma$ ([tis], ‘them’-GEN&ACC.FEM). Since they can both be genitive, whenever I used it in a sentence, it would give me 2 solutions, one for Singular and one for Plural, a problem which wouldn’t occur if I could use the greek alphabet on XLE. In the beginning I decided to leave it like this and explain in my report the problem but I eventually decide to not follow the phonetic transcription for this specific clitic but write the singular one as ‘ths’ because the letter ‘h’ is used as the greek letter ‘η’ [i] when we write in Greeklish (Greek language written in Latin script).

Lastly, one problem for which I did not find a solution is that the third person clitic pronouns are exactly the same with the definite determiners but I could not include them under the same lexical entry because they are different grammatical categories. Whenever I tried to include both of them in the grammar, it would give me a duplicate message and zero solutions for the the parsing sentence. As a result I decided to comment out the genitive and accusative determiners of all the genders. This might grow concerns about the construction of the PPs, but it is not a problem in Modern Greek. When the noun of a PP is definite, the preposition *σε* ([se], 'at/to') gets fused with the definite determiner, so if we want to say 'at the park' (definite), we say *στο πάρκο* [sto `parko]. But if we want to say 'at a park'(indefinite), we say *σε ένα πάρκο* [se `ena `parko] and the preposition appears plain as it is followed by the indefinite article.

Appendix: Glossed Testsuite

The examples are given with a Modern Greek orthography and followed by their phonetic transcription according to the IPA with each word's stress marks. Glossing follows the Leipzig Glossing Rules with each word being divided in its morphemes.

Abbreviations

1, 2, 3	1st / 2nd / 3rd person
SG, PL	singular / plural
NOM, ACC, GEN, VOC	nominative / accusative / genitive / vocative
M, F, N	masculine / feminine / neuter
DEF, INDEF	definite / indefinite article
CL	clitic pronoun
PRS, PST	present / past tense
IPFV, PFV	imperfective / perfective aspect
COMP	complementizer ('that')

Examples

(1)

η	Ελλάδα-α	έχ-ει	πολλ-ούς	ξέν-ους	τουρίστ-ες
<i>i</i>	<i>Eˈlaða-a</i>	<i>ˈeç-i</i>	<i>po ˈl-us</i>	<i>ˈksen-us</i>	<i>tu ˈrist-es</i>
DEF.NOM.SG.F	Greece-NOM.SG.F	have-3SG.PRS	many-ACC.PL.M	foreign-ACC.PL.M	tourist-ACC.PL.M

‘Greece has many foreign tourists.’

Notes: Grammatical. Determiner, adjective and noun agree in gender/case/number(M.ACC.PL): 1 solution.

(2)

η	Ελλάδα-α	έχ-ει	πολλ-ούς	ξέν-οι	τουρίστ-ες
<i>i</i>	<i>Eˈlaða-a</i>	<i>ˈeç-i</i>	<i>po ˈl-us</i>	<i>ˈksen-i</i>	<i>tu ˈrist-es</i>
DEF.NOM.SG.F	Greece-NOM.SG.F	have-3SG.PRS	many-ACC.PL.M	foreign-NOM/ VOC.PL.M	tourist-ACC.PL.M

‘Greece has many foreign tourists.’

Notes: Ungrammatical (it is not visible in English). The suffix [-i] of the adjective [ˈkseni] is only lexically specified for NOM.PL or VOC.SG, never ACC. So it does not satisfy the determiner-adjective-noun case agreement: 0 solutions.

(3)

η	Ελλάδα-α	έχ-ει	πολλ-ών	ξέν-ων	τουριστών
<i>i</i>	<i>E`lað-a</i>	<i>`eç-i</i>	<i>po`l-on</i>	<i>`ksen-on</i>	<i>turi`st-on</i>
DEF.NOM.SG.F	Greece-NOM.SG.F	have-3SG.PRS	many-GEN.PL.M	foreign-GEN.PL.M	tourist-GEN.PL.M

‘Greece has many foreign tourists.’

Notes: Ungrammatical (it is not visible in English). Determiner, adjective and noun agree internally (GEN.PL.M), but the VP object slot only accepts an accusative NP. So the genitive object is never licensed: 0 solutions

(4)

σε	αγαπά-ω	πολύ
<i>se</i>	<i>aga`pa-o</i>	<i>po`li</i>
2SG.ACC.CL	love-PRS.1SG.IPFV	very

‘I love you very much.’

Notes: Grammatical. The object clitic is in accusative case: 1 solution.

(5)

σου	αγαπά-ω	πολύ
<i>su</i>	<i>aga`pa-o</i>	<i>poli</i>
2SG.GEN.CL	love-PRS.1SG.IPFV	very

‘I love you very much.’

Notes: Ungrammatical (it is not visible in English). The clitic [su] is lexically genitive, but [aga`pao] (TRANS) licenses only an accusative OBJ, never OBJ2. So the clitic case and the verb frame do not match: 0 solutions

(6)

έ-δω-σ-α	στ-ην	Μαρί-α	έν-α	βιβλί-ο	σήμερα
<i>`e-do-s-a</i>	<i>st-in</i>	<i>Ma`ri-a</i>	<i>`en-a</i>	<i>vi`vli-o</i>	<i>`simera</i>
PST-give-PFV-1SG	to_the-ACC.SG.F	Maria-ACC.SG.F	INDEF.ACC.SG.N	book-ACC.SG.N	today

‘I gave to Maria a book today.’

Notes: Grammatical. Recipient realized as an OBL-TO (DITRANS's OBJ+OBL-TO branch): 1 solution.

(7)

σήμερα	τ-ης	έ-δω-σ-α	έν-α	βιβλί-ο
`simera	<i>t-hs [tis]</i>	`e-do-s-a	`en-a	vi`vli-o
today	3PL.GEN.F.CL	PST-give-PFV-1SG	INDEF.ACC.SG.N	book-ACC.SG.N

‘Today I gave her a book.’

Notes: Grammatical. Recipient realized as the genitive clitic OBJ2 (DITRANS's OBJ+OBJ2 branch). Also, adverb fronting: 1 solution.

(8)

σήμερα	τ-ης	έ-δω-σ-α	μια	βιβλί-ο
`simera	<i>t-hs [tis]</i>	`e-do-s-a	<i>mja</i>	vi`vli-o
today	3PL.GEN.F.CL	PST-give-PFV-1SG	INDEF.ACC.SG.F	book-ACC.SG.N

‘Today I gave her a book.’

Notes: Ungrammatical (it is not visible in English). The indefinite determiner [mja] (feminine) does not agree with the noun [vi`vlio] (neuter) inside the object NP: 0 solutions.

(9)

σου	είπ-ε	ότι	αγαπά-ει	μια	όμορφ-η	κοπέλ-α
<i>su</i>	`ip-e	`oti	aga`pa-i	<i>mja</i>	`omorf-i	ko`pel-a
2SG.GEN.CL	tell.3SG.PST.PF V	COMP	love-3SG.PRS.I PFV	INDEF.ACC.S G.F	beautiful-ACC.SG.F	girl-ACC.SG.F

‘He/she told you that he/she loves a beautiful girl.’

Notes: Grammatical. Both clauses have a pro-drop subject which is the same (3SG). The argument frame of the verb [‘ipe] is GEN-COMP, which means that it takes one OBJ2 and a COMP: 1 solution.

(10)

μου	είπ-αν	ότι	πήγ-αν	στ-ον	γιατρ-ό
<i>mu</i>	`ip-an	`oti	`pig-an	<i>st-on</i>	<i>ja`tr-o</i>
1SG.GEN.CL	tell-3PL.PST.PFV	COMP	go-3PL.PST.PFV	to_the-ACC.SG.M	doctor-ACC.SG.M

‘They told me that they went to the doctor.’

Notes: Grammatical. Same as the previous sentence but without an OBJ in the COMP and with a PP: 1 solution.

(11)

σας	είπ-α	ότι	πριν	μια	εβδομάδ-α	πήγ-αμε	στ-ον	κινηματογράφ-ο	στ-ο	κέντρ-ο
<i>sas</i>	<i>`ip-a</i>	<i>`oti</i>	<i>prin</i>	<i>mja</i>	<i>evdo`ma d-a</i>	<i>`pig-ame</i>	<i>st-on</i>	<i>kinimato`graf-o</i>	<i>st-o</i>	<i>`kedr-o</i>
2PL.GE N.CL	tell-1SG. PST.PFV	COMP	before	INDEF.A CC.SG.F	week- ACC.SG. F	go-1PL.P ST.PFV	to_the- ACC.SG. M	cinema- ACC.SG. M	in_the- ACC.SG. N	center- ACC.SG. N

'I told you that a week ago we went to the cinema in the centre.'

Notes: Grammatical. Same as the previous sentence but with an AdvP in the COMP and two consecutive PPs: 1 solution.

(12)

τ-ης	είπ-α	ότι	τ-ην	ζητά-νε	στ-ο	τηλέφων-ο
<i>t-hs [tis]</i>	<i>`ip-a</i>	<i>`oti</i>	<i>t-in</i>	<i>zi`ta-ne</i>	<i>st-o</i>	<i>ti`lefon-o</i>
3SG.GEN.F.CL	tell-1SG.PST. PFV	COMP	3SG.ACC.F.CL	ask_for-3PL. PRS.IPFV	on_the- ACC.SG.N	phone- ACC.SG.N

'I told her that they ask for her on the phone.'

Notes: Grammatical. [sto] attaches as a loose ADJUNCT (PTYPE=sem, 'on') rather than a governed OBL-TO, which happens to give exactly the right English idiom 'on the phone': 1 solution.

(13)

τ-ην	είπ-α	ότι	τ-ην	ζητάνε	στ-ο	τηλέφων-ο
<i>t-in</i>	<i>`ip-a</i>	<i>`oti</i>	<i>t-in</i>	<i>zi`ta-ne</i>	<i>st-o</i>	<i>ti`lefon-o</i>
3SG.ACC.F.CL	tell-1SG.PST. PFV	COMP	3SG.ACC.F.CL	ask_for-3PL. PRS.IPFV	on_the- ACC.SG.N	phone- ACC.SG.N

'I told her that they ask for her on the phone.'

Notes: Ungrammatical (it is not visible in English). The argument frame of the verb [ipa] in this sentence is GEN-COMP, which means that it takes one OBJ2 and a COMP but the clitic [tin] in the main clause is in accusative case. So it cannot fill the OBJ2 spot: 0 solutions.

Regarding the use of the accusative for the OBJ2 clitic pronoun instead of the genitive, it should be noted that this phenomenon, although it does not agree with the syntax of Standard modern Greek, is observed in areas of Northern Greece. It is a frequently mentioned feature when someone imitates the northern dialect, particularly that of the city I come from, Thessaloniki.

(14)

τ-ους	ενημέρω-σ-αν	ότι	φτάν-ουν	αργά	στ-ο	λιμάν-ι
<i>t-us</i>	<i>eni`mero-s-an</i>	<i>`oti</i>	<i>`ftan-un</i>	<i>ar`ga</i>	<i>st-o</i>	<i>li`man-i</i>
3PL.ACC.M.CL	inform-PFV-3PL.PST	COMP	arrive-3PL.PRS. IPFV	late	at_the-ACC.SG.N	port-ACC.SG.N

‘They informed them that they arrive late at the port.’

Notes: Grammatical. The verb [eni`erosan] in Modern greek can take as object arguments an OBJ followed by a COMP, this is the reason I created the template ‘OBJ-COMP’. So its clitic pronoun is in accusative: 1 solution.

(15)

μ-ας	ανακοινώ-σ-ατε	ότι	μετακομί-ζ-ουν	στ-ο	εξωτερικ-ο	σύντομα
<i>m-as</i>	<i>anaki`no-s-ate</i>	<i>`oti</i>	<i>metako`mi-z-oun</i>	<i>st-o</i>	<i>eksoteri`k-o</i>	<i>`sidoma</i>
1PL.GEN.CL	announce-2PL. PST.PFV	COMP	move-3PL.PRS. IPFV	to_the-ACC.SG.N	abroad-ACC.SG.N	soon

‘You announced to us that they are moving abroad soon.’

Notes: Grammatical. Both verbs have pro-drop subjects, which are different: 1 solution.

(16)

σ-ου	υπενθυμί-ζ-ω	ότι	αύριο	γρά-φ-εις	έν-α	πολύ	δύσκολ-ο	διαγώνισμ-α
<i>s-u</i>	<i>ipenθi`mi-z-o</i>	<i>`oti</i>	<i>`avrio</i>	<i>`gra-f-is</i>	<i>`en-a</i>	<i>po`li</i>	<i>`diskol-o</i>	<i>dia`gonizm-a</i>
2SG.GEN. CL	remind-1SG .PRS.IPFV	COMP	tomorrow	write-IPFV.2SG.P RS	INDEF. ACC.SG.N	very	difficult-ACC.SG.N	test-ACC.SG.N

‘I remind you that tomorrow you write a very difficult test.’

Notes: Grammatical. In the embedded clause we have a fronted adverb and an NP as the object, which has a determiner, an adverb and an adjective: 1 solution.

(17)

τ-ης	τ-ο	είπ-α
<i>t-hs [tis]</i>	<i>t-o</i>	<i>`ip-a</i>
3PL.GEN.F.CL	3PL.ACC.N.CL	tell-1SG.PST.PFV

‘I told her that.’

Notes: Grammatical. The verb has OBJ and OBJ2. The OBJ2 always precedes the OBJ when they occur together: 1 solution.

(18)

τ-ο	τ-ης	είπ-α
<i>t-o</i>	<i>t-hs [tis]</i>	<i>`ip-a</i>
3PL.ACC.N.CL	3PL.GEN.F.CL	tell-1SG.PST.PFV

‘I told that her.’

Notes: Ungrammatical. It does not apply to the strict positioning of the clitics as OBJ and OBJ2: 0 solutions.

(19)

τ-ης	είπ-α	ότι	ο	Νίκος	τ-ην	αγαπά-ει
<i>t-hs [tis]</i>	<i>`ip-a</i>	<i>`oti</i>	ο	<i>`Nik-os</i>	t-in	aga`pa-i
3PL.GEN.F.CL	tell-1SG.PST. PFV	COMP	DEF.NOM.SG. M	Nik- NOM.SG.M	3PL.ACC.F.CL	love- PRS.3SG.IPFV

‘I told her that Nikos loves her.’

Notes: Grammatical. An example to show that the embedded clause can also have an overt subject and not only a pro-drop: 1 solution.

Bibliography

Philippaki-Warbuton, I. (1987). The theory of empty categories and the pro-drop parameter in Modern Greek. *Journal of Linguistics*, 23(2), 289–318. [doi:10.1017/S0022226700011282](https://doi.org/10.1017/S0022226700011282)